

Och

1 Introduction

Industry relies on an enormous amount of runtime checks to analyse the correctness of the software they produce. This largely takes the form of test suites, human QA testing, or collecting bug reports from users.

Static analysis is a better way of catching bugs than runtime checks when you can use it, as it notifies developers of mistakes earlier in the development pipeline, making the fixes quicker and more accurate. In the case of bugs which would have gone all the way to production, it can also make the end product better.

So why not catch *every* bug at compile time? It would mean developers never need to write or run tests, and never need to fix bugs because the bugs were never introduced. We have the tools to prove full functional correctness of software, so why don't we use them?

Industry has adopted many forms of static analysis, but only when the economics of it work out. For static analysis to be worthwhile, the time developers spend engaging with static analysis tools must be paid for by:

- Speeding up the rest of the development pipeline (e.g. by removing the need to write a test or fix a bug)
- Making the end product better (e.g. by making the software less buggy and more secure)

The goal for my PhD is to bring the strong type systems developed by academia to the masses by making a pleasant, useful language which addresses industry pain points.

The pinnacle of type systems research for me is allowing developers to prove the functional correctness of their programs via formal verification. In industry enormous amounts of developer hours are wasted catching and fixing bugs, and generally dealing with the consequences of software which doesn't *quite* work. I hope to stop these hours from being wasted, and improve the standard and speed of software engineering.

In my opinion the most principled and ergonomic way of enabling developers to formally verify their code is by having them write software directly in a language which natively supports dependent types. There are other ways to verify the correctness of your software across all inputs without running it, which I explain in further detail in Section XYZ.

The AI revolution has created the perfect environment for a language like Ochre to succeed:

- **Security vulnerabilities caused by unverified software are costing more than ever before:** due to a recent increase in the intelligence of LLMs, coding agents have been finding enormous quantities of security vulnerabilities in foundational software: Firefox found and fixed 400 security bugs in April 2026 compared to the usual 20, a 9-year dormant linux kernel bug was found which allowed root-level privilege escalation, and Anthropic has refused to release their most recent model due to concerns bad actors would use it for hacking.
- **Coding agents have brought the cost of software engineering down significantly** allowing developers to be much more ambitious. As an example: Bun, a major open source project, was completely re-written from Zig to Rust in 9 days with a single million-line-of-code PR which has been merged. The bottleneck to software engineering is no longer intelligence and time, it's how good you are at specifying what you want. This both decreases the barrier to entry for a new language like Ochre, but also increases the reward because verification is the most precise way imaginable of specifying what one wants.

After playing around with dependent types in an imperative programming setting during the masters thesis, I came to the conclusion that a fairly uncommon combination of language features would lead to an excellent combination of developer ergonomics and expressivity.

I want to study the most unweildly of these features in isolation with a core calculus before building Ochre, specifically:

- The “terms are types” philosophy, as exonerated by PSS (Hutchins 2010). In Och this comes out roughly to “types are terms with holes”.
- Terms/types form a subtyping lattice, where supertypes can be freely substituted for any subtype without any coercions.

1.1 Methodology

Och has been designed in an experimental fashion, where a set of goal programs drove the design of the typing rules. As the typing rules succeeded with a given set of goal programs, more goal programs were made and the iteration restarted

As a result of this

- Och is lacking a satisfying unifying theory and attempts of the soundness proof have been unsuccessful. This makes for a natural next step now that I have a set of rules which “mostly work”.
- There is quite an impressive set of test cases (see Section 5) which demonstrate that this idea has impressive expressivity both in terms of the complexity of the programs it is able to accept, and the subtlety of the bugs it is able to catch. The most succinct answer to “why Och” is that I wanted these example programs to be handled correctly, and no system I could find does it quite to my liking.

As a taster of what is to come: using Church/Scott encodings I have expressed booleans, natural numbers, finite sets ($\text{Fin } n$), pairs, lists, lists with compile-time-known-length, and safe list indexing.

These encodings use dependent elimination throughout, allowing the type system to catch bugs as subtle as adding the *wrong* two numbers, instead of just checking that the addition was well-shaped.

1.2 Goals

Soundness - Ochre is only valuable if programmers can trust the compiler to catch runtime errors. Proving the soundness of systems with the above features, let alone ownership + mutation is novel research as far as I can tell, Och is a place to do that research.

Expressivity - The type system must be able to articulate properties and catch bugs that people care about. E.g. it must be expressive enough to tell the difference between $x + y$ and $x + x$ if one is correct and the other incorrect.

Extensibility - Och is the first in a series of languages which will eventually culminate in Ochre. As such, there is no point working with features which will not be re-usable. This has already manifested in the decision to tackle arbitrary unbounded recursion via `fix` instead of adding primitive inductive types and only well-founded induction.

2 Related Work

Pure Subtype Systems (Hutchins 2010) - Och was mostly developed before I found PSS, and to my delight PSS is an excellent execution on my core ideas, so I am in the process of re-framing Och as an *extension* of PSS instead of a new system. PSS explores the idea that “terms and types are the same thing” not just in terms of syntax but also semantics. The PSS paper is quite short, very easy to understand, and strongly recommended reading for anyone engaging with Och.

3 Language Semantics

This section describes what each piece of the language does in natural language to give the reader an intuition for the objects involved before we get into the hard typing rules examples and metatheory.

Och syntax:

$e, \tau ::= x$	variable
$\lambda(x \leq \tau).e$	lambda
$e_1 e_2$	application
$\top$	universe (top)
$\perp$	primitive bottom
$\iota(x \leq \tau).e$	self-type binder
$\text{fix}(x \leq \tau).e$	recursive binder

There is no separate syntactic category for types: every τ above is itself a term.

Variables, lambdas, application as per the standard λ -calculus, except λ has a domain type annotation which is respected by the type checker.

$\top$ **and** $\perp$ are the top and bottom of the subtyping lattice. $\top$ represents the widest possible type, which tells you nothing about the term being typed, and $\perp$ is the empty type.

Fix $\text{fix}(x \leq \tau).e$ allows a term to be defined in terms of itself. x is the reference to the whole fix expression itself, and τ is the “upper bound” which the whole expression can be assumed to have while it’s being type checked (which prevents loops during type checking).

Iota $\iota(x \leq \tau).e$ allows the definition of a type to include a reference to *the term inhabiting it*. This means $e \sqsubseteq \iota(x : \tau_0).\tau_1$ is reduced to $e \sqsubseteq \tau_1[x \mapsto e]$ during checking (notice: the LHS has moved to the RHS). This is a Cedille-style self type (Fu and Stump 2014), and allows for church encodings with dependent elimination instead of having to add inductive datatypes to the language as a primitive.

3.1 Runtime semantics

You may think the runtime semantics of Och are unimportant since this is type systems research, but with how I’ve laid out the soundness proofs $\Downarrow$ plays a crucial role: soundness states “if a program type checks, it will succeed at runtime”, therefore $\Downarrow$ must *reject* ill-formed programs, otherwise our soundness becomes vacuously solvable (true on the RHS of an implication).

The concrete evaluator is a substitution-based call-by-value big-step interpreter on closed terms. $\{\lambda, \top, \perp\}$ are the only values. fix and ι eagerly unroll by substituting their self-reference into the body. Application is pure β -reduction — only lambdas can be applied.

Och has no concept of levels/universes/stages, so the type checker cannot enforce that $\{\top, \perp, \iota\}$ don’t go into runtime, so runtime needs to handle them gracefully. Adding universes (see §3.6 of PSS (Hutchins 2010)) would let us erase type-level arguments during compilation, after which $\{\top, \perp, \iota\}$ would never appear at level 0.

We write the judgment $e \Downarrow v$ for “closed term e concretely evaluates to value v ”. Note: there is no context and no free/abstract variables: everything is eagerly substituted in.

$$\begin{array}{c}
\frac{}{v \Downarrow v} \text{E-VAL} \qquad \frac{f \Downarrow \lambda(x \leq \tau).b \quad a \Downarrow v_a \quad b[x \mapsto v_a] \Downarrow v}{f a \Downarrow v} \text{E-APP} \\
\\
\frac{b[x \mapsto \iota(x \leq \tau).b] \Downarrow v}{\iota(x \leq \tau).b \Downarrow v} \text{E-IOTA} \qquad \frac{b[x \mapsto \text{fix}(x \leq \tau).b] \Downarrow v}{\text{fix}(x \leq \tau).b \Downarrow v} \text{E-FIX} \\
\\
\text{where } v \in \text{Value} ::= \lambda(x \leq \tau).e \qquad \text{lambda} \\
\qquad \qquad \qquad | \perp \qquad \text{primitive bottom} \\
\qquad \qquad \qquad | \top \qquad \text{universe (top)}
\end{array}$$

The most important aspect of the runtime semantics are what is *missing* from the runtime semantics:

- Free variables are not values and have no rule, therefore they cannot occur at runtime, which is what forces the type system to rule out ill-scoped variables.

- Application can only apply to lambdas, which is what forces the type system to verify applications are well typed w.r.t. the function domain.

Fix and iota eagerly unroll — a program like $\text{fix}(x \leq \top).x$ loops forever, which is the correct behaviour for non-terminating recursion. This lines up with what a programmer would expect to happen if they wrote `while true {}` in Rust or a lesser programmer would expect if they wrote `let x = x in x` in Haskell.

4 Typing Rules

Och’s type system is structured around two judgements:

Well-formedness $\Gamma \vdash e \text{ wf}$ states “ e is well-formed under Γ ” and is the “entry point” for the type system. Defined in Section 4.2. Delegates all type-comparison questions to subtyping.

Subtyping $S; \Gamma \vdash a \sqsubseteq b$ states “ a is a subtype of b under Γ , assuming coinductive hypotheses S ”. Subtyping in Och means set inclusion: $A \sqsubseteq B$ iff every value of A is also a value of B . Section 4.3 explains the S coinductive hypothesis system.

4.1 Context Γ

Γ is the typing context defined as $\Gamma ::= \emptyset \mid \Gamma, x \leq a$. Lookup is $\Gamma(x)$. Each entry describes what the type checker knows about a variable statically.

4.2 Well-formedness

The well-formedness judgment $\Gamma \vdash e \text{ wf}$ determines whether term e is well-formed under context Γ . In other systems this role would be served by a $\Gamma \vdash a : \tau$ rule which assigns a type to a term, but there is no need to assign a type to terms in Och because every term is already trivially its own type by [S-Refl].

Well-formedness validates purely structural conditions: annotations are types, binders are in scope, and applications have Π -typed functions with domain-inhabiting arguments.

$$\begin{array}{c}
\frac{x \in \text{dom}(\Gamma)}{\Gamma \vdash x \text{ wf}} \text{W-VAR} \qquad \frac{}{\Gamma \vdash \top \text{ wf}} \text{W-TYPE} \qquad \frac{}{\Gamma \vdash \perp \text{ wf}} \text{W-BOT} \\
\\
\frac{\Gamma \vdash A \text{ wf} \quad \Gamma, x \leq A \vdash b \text{ wf}}{\Gamma \vdash \lambda(x \leq A).b \text{ wf}} \text{W-LAM} \qquad \frac{\Gamma \vdash f \text{ wf} \quad \Gamma \vdash a \text{ wf} \quad \emptyset; \Gamma \vdash f \sqsubseteq \lambda(x \leq A).B \quad \emptyset; \Gamma \vdash a \sqsubseteq A}{\Gamma \vdash f a \text{ wf}} \text{W-APP} \\
\\
\frac{\Gamma \vdash A \text{ wf} \quad \Gamma, x \leq A \vdash b \text{ wf}}{\Gamma \vdash \iota(x \leq A).b \text{ wf}} \text{W-IOTA} \qquad \frac{\Gamma \vdash A \text{ wf} \quad \Gamma, x \leq A \vdash b \text{ wf}}{\Gamma \vdash \text{fix}(x \leq A).b \text{ wf}} \text{W-FIX}
\end{array}$$

[W-Var] requires variables be well-scoped.

[W-Lam], [W-Iota], [W-Fix] check that the annotation is well-formed and that the body is well-formed under the extended context.

[W-App] is the only rule with subtyping premises: the function must subtype some $\lambda(x \leq A).B$ (it has a Π -type), and the argument must subtype the domain A . No type is assigned to the result — the result’s type is the result itself (by [S-Refl]), and if the caller needs to know it subtypes something, that is a separate subtyping question.

4.3 Seen set S

Due to the prevalent usage of unbounded recursion in type definitions and encodings, many subtype checks involve infinite proof trees (e.g. $\emptyset; \emptyset \vdash \text{add one two} \sqsubseteq \text{Nat}$).

To derive these infinite trees in finite time Claude said I should use “the Brandt-Henglein device for equirecursive subtyping” as used by Brandt and Henglein (1998). Claude explains it as: “ S is a finite representation of a coinductive (potentially infinite) proof tree: when a goal recurs, the branch closes rather than recursing forever, encoding a regular infinite derivation as a finite one.”

As far as I can tell the reason it is remarkable enough to have a name is not that using it is particularly hard, but that convincing oneself it is a sound technique is hard, and Brandt/Henglein did just that. I

am informed it only works when the cycles it closes contain at least one “productive” step, which are grouped together into Section 4.7.

The “real” subtyping judgment is $\emptyset; \Gamma \vdash a \sqsubseteq b$ (empty hypothesis set); non-empty S arises only inside a derivation.

4.4 Subtyping Rules

The subtyping rules fall into four categories. Each serves a distinct purpose; knowing which category a rule belongs to predicts its shape.

Category	Purpose	Rules	Extends S ?
Structural (Section 4.5)	Plumbing: recurse without inspecting constructors on either side	S-Refl, S-Top, S-BotL, S-Trans, S-Hyp, S-Var	no
Congruence (Section 4.6)	Match constructor on both sides; reduce to sub-obligations with variance	S-Lam, S-App-Cong, S-Iota-Cong, S-Fix-Cong	no
Productive unfolding (Section 4.7)	Unfold a recursive binder (ι /fix); extend S ; enable coinductive closure	S-Iota-Intro, S-Unfold-Iota-L/R, S-Unfold-Fix-L/R	yes
Conversion (Section 4.8)	Close under head reduction so subtyping is closed under computation	S-Beta-L/R	no

Table 1: Rule taxonomy for declarative subtyping

The S -extension column matters: S-Hyp can only fire against entries that some ancestor productive rule installed, so any path to S-Hyp must cross an unfold — the productivity requirement made mechanical.

4.5 Structural rules

“Structural” rules talk about $\sqsubseteq$ as a relation on terms without inspecting either side’s head constructor: reflexivity, transitivity, the top element, hypothesis lookup, and variable lookup.

$$\begin{array}{ccc}
\frac{}{S; \Gamma \vdash e \sqsubseteq e} \text{S-REFL} & \frac{S; \Gamma \vdash a \sqsubseteq b \quad S; \Gamma \vdash b \sqsubseteq c}{S; \Gamma \vdash a \sqsubseteq c} \text{S-TRANS} & \frac{(a, b) \in S}{S; \Gamma \vdash a \sqsubseteq b} \text{S-HYP} \\
\frac{}{S; \Gamma \vdash x \sqsubseteq \Gamma(x)} \text{S-VAR} & \frac{}{S; \Gamma \vdash e \sqsubseteq \top} \text{S-TOP} & \frac{}{S; \Gamma \vdash \perp \sqsubseteq e} \text{S-BOTL}
\end{array}$$

Ex falso via subsumption. There is no dedicated “absurd” eliminator. If $a \sqsubseteq \perp$ is derivable, then $a \sqsubseteq e$ for every e via [S-Trans] on [S-BotL]. The “contradiction” discharge is subsumption alone. This matches the DOT (Amin et al. 2012) tradition: $\perp$ inhabits every type trivially in subtyping, so any term whose type is already $\perp$ flows into any expected type without further ceremony.

[S-Trans] is a constructor, not a derived theorem. Given the interpretation of types as sets of terms, and subtyping as subset over those sets, one would expect transitivity to be provable instead of axiomatic. Eliminating transitivity as a primitive rule (*transitivity elimination*) is highly desirable: it simplifies metatheory and makes the relation syntax-directed. PSS (Hutchins 2010) — a closely related system — did not manage to eliminate transitivity. Pasquale and García-Pérez (Pasquale and García-Pérez 2026) partially succeeded in a continuation of that work, but required switching the entire theory to a Krivine machine.

I have no satisfying answer to whether or not transitivity is eliminatable in Och, but I will try get one.

4.6 Congruence rules

Congruence rules do inspect the head constructor on both sides, *require it to match*, and reduce the goal to sub-obligations on the immediate sub-terms with appropriate variance.

$$\begin{array}{c}
\frac{S; \Gamma \vdash A_2 \sqsubseteq A_1}{S; \Gamma, x \leq A_2 \vdash b_1 \sqsubseteq b_2} \text{S-LAM-CONG} \quad \frac{S; \Gamma \vdash f_2 \sqsubseteq f_1 \quad S; \Gamma \vdash a_2 \sqsubseteq a_1}{S; \Gamma \vdash a_1 \sqsubseteq a_2} \text{S-APP-CONG} \\
\frac{S; \Gamma \vdash A_1 \sqsubseteq A_2}{S; \Gamma, x \leq A_2 \vdash B_1 \sqsubseteq B_2} \text{S-IOTA-CONG} \quad \frac{S; \Gamma \vdash A_1 \sqsubseteq A_2}{S; \Gamma, x \leq A_2 \vdash b_1 \sqsubseteq b_2} \text{S-FIX-CONG} \\
\frac{S; \Gamma, x \leq A_2 \vdash B_1 \sqsubseteq B_2}{S; \Gamma \vdash \iota(x \leq A_1).B_1 \sqsubseteq \iota(x \leq A_2).B_2} \text{S-IOTA-CONG} \quad \frac{S; \Gamma, x \leq A_2 \vdash b_1 \sqsubseteq b_2}{S; \Gamma \vdash \text{fix}(x \leq A_1).b_1 \sqsubseteq \text{fix}(x \leq A_2).b_2} \text{S-FIX-CONG}
\end{array}$$

The equivalence premise on [S-App-Cong] (both directions on the argument) is necessary because a neutral head can use its argument at any variance, so equivalence is the only sound congruence.

4.7 Productive unfolding (extend S)

A rule is *productive* when it replaces a goal whose head is a recursive binder (ι or fix) with a goal where that binder has been unfolded once. Unfolding makes the term *larger* in syntactic size, so no structural induction can traverse an arbitrary chain of unfolds. Productivity instead provides a *coinductive* handle: once at least one unfold has fired, the original goal is guaranteed to eventually re-appear as an ancestor, at which point [S-Hyp] can close the derivation.

The rules below are the only ones that extend S .

Let $S' = S, (a, b)$ where $S; \Gamma \vdash a \sqsubseteq b$ is the current goal.

$$\begin{array}{c}
\frac{S'; \Gamma \vdash a \sqsubseteq A}{S'; \Gamma \vdash a \sqsubseteq B[x \mapsto \iota(x \leq A).B]} \text{S-IOTA-INTRO} \quad \frac{S'; \Gamma \vdash B[x \mapsto \iota(x \leq A).B] \sqsubseteq c}{S; \Gamma \vdash \iota(x \leq A).B \sqsubseteq c} \text{S-UNFOLD-IOTA-L} \\
\frac{S'; \Gamma \vdash a \sqsubseteq B[x \mapsto \iota(x \leq A).B]}{S; \Gamma \vdash a \sqsubseteq \iota(x \leq A).B} \text{S-UNFOLD-IOTA-R} \\
\frac{S'; \Gamma \vdash b[x \mapsto \text{fix}(x \leq A).b] \sqsubseteq c}{S; \Gamma \vdash \text{fix}(x \leq A).b \sqsubseteq c} \text{S-UNFOLD-FIX-L} \quad \frac{S'; \Gamma \vdash a \sqsubseteq b[x \mapsto \text{fix}(x \leq A).b]}{S; \Gamma \vdash a \sqsubseteq \text{fix}(x \leq A).b} \text{S-UNFOLD-FIX-R}
\end{array}$$

S-Unfold-Iota-R is the weaker sibling of S-Iota-Intro (same conclusion, no annotation premise), needed to close the equivalence of ι -unfolding.

S-Iota-Intro is **the** rule which allows dependent elimination: it puts the thing being typed **into** the type, allowing the type to depend on the inhabitant.

Worked S-Iota-Intro example: $\text{true} \sqsubseteq \text{DBool}$. With

$$\begin{aligned}
\text{true} &:= \lambda(P \leq \top). \lambda(t \leq \top). \lambda(f \leq \top). t \\
\text{false} &:= \lambda(P \leq \top). \lambda(t \leq \top). \lambda(f \leq \top). f \\
\text{DBool} &:= \text{fix}(B \leq \top). \iota(\text{self} \leq B). \lambda(P \leq B \rightarrow \top). \lambda(t \leq P \text{ true}). \lambda(f \leq P \text{ false}). P \text{ self}
\end{aligned}$$

the constructors are plain lambdas with $\top$ domains — they carry no recursive reference to DBool . The subtyping check $\emptyset; \Gamma \vdash \text{true} \sqsubseteq \text{DBool}$ still requires the coinductive seen-set to close, because [S-Iota-Intro]’s annotation premise cycles back to the root goal:

$$\begin{array}{l}
\emptyset; \Gamma \vdash \text{true} \sqsubseteq \text{DBool} \quad \text{root goal} \\
S_1; \Gamma \vdash \text{true} \sqsubseteq \iota(\text{self} \leq \text{DBool}). \lambda(P \leq \text{DBool} \rightarrow \top). \lambda(t \leq \\
P \text{ true}). \lambda(f \leq P \text{ false}). P \text{ self} \quad \text{S-Unfold-Fix-R} \\
\text{where } S_1 = \{(\text{true}, \text{DBool})\} \\
\text{S-Iota-Intro — annotation premise:} \\
S_1; \Gamma \vdash \text{true} \sqsubseteq \text{DBool} \quad \text{S-Hyp } \checkmark \\
\text{root goal reappears — closed by } (\text{true}, \text{DBool}) \in S_1 \\
\text{S-Iota-Intro — body premise (after } [\text{self} \mapsto \text{true}]):
\end{array}$$

$$\begin{array}{ll}
S_1; \Gamma \vdash \text{true} \sqsubseteq \lambda(P \leq \text{DBool} \rightarrow \top). \lambda(t \leq P \text{ true}). \lambda(f \leq P \text{ false}). P \text{ true} & \text{S-Lam} \\
\text{contravariant: } (\text{DBool} \rightarrow \top) \sqsubseteq \top \text{ by S-Top } \checkmark & \\
\text{covariant body under } P \leq \text{DBool} \rightarrow \top: & \\
S_1; \Gamma \vdash \lambda(t \leq \top). \lambda(f \leq \top). t \sqsubseteq \lambda(t \leq P \text{ true}). \lambda(f \leq P \text{ false}). P \text{ true} & \text{S-Lam} \\
\text{contravariant: } P \text{ true} \sqsubseteq \top \text{ by S-Top } \checkmark & \\
\text{covariant body under } t \leq P \text{ true: S-Lam again} & \\
\text{contravariant: } P \text{ false} \sqsubseteq \top \text{ by S-Top } \checkmark & \\
\text{covariant body: } t \sqsubseteq P \text{ true by S-Var } (t \leq P \text{ true} \in \Gamma) \checkmark &
\end{array}$$

Because the constructors have $\top$ domains, every contravariant [S-Lam] premise is discharged by [S-Top]. The only coinductive step is the [S-Iota-Intro] annotation premise, which closes via [S-Hyp] after one productive unfold. This is the Brandt–Henglein discipline in action: recursion in the subtyping judgment is legal *only across a productive unfold*, so non-productive loops (e.g. reflexivity-by-loop) cannot sneak through.

4.8 Conversion

The subtyping relation must be closed under head reduction: if a computes to a' , then $a \sqsubseteq b$ should hold iff $a' \sqsubseteq b$. These rules provide that closure.

$$\frac{S; \Gamma \vdash \text{body}[x \mapsto \text{arg}] \sqsubseteq b}{S; \Gamma \vdash (\lambda(x \leq A). \text{body}) \text{ arg} \sqsubseteq b} \text{S-BETA-L} \qquad \frac{S; \Gamma \vdash a \sqsubseteq \text{body}[x \mapsto \text{arg}]}{S; \Gamma \vdash a \sqsubseteq (\lambda(x \leq A). \text{body}) \text{ arg}} \text{S-BETA-R}$$

5 Example Programs

All encodings below are Church/Scott-style: data types are represented as their own eliminators, with no primitive inductive types in the language. Every definition is mechanised and tested in Lean 4 (see Section 7). This section builds from simple encodings to two programs that demonstrate Och’s ability to catch subtle bugs in dependently-typed code.

5.1 Booleans

$$\begin{aligned}
\text{true} &:= \lambda(P \leq \top). \lambda(t \leq \top). \lambda(f \leq \top). t \\
\text{false} &:= \lambda(P \leq \top). \lambda(t \leq \top). \lambda(f \leq \top). f \\
\text{Bool} &:= \lambda(X \leq \top). \lambda(t \leq X). \lambda(f \leq X). X
\end{aligned}$$

Bool is a standard Church-encoded boolean: the eliminator takes two branches and a return type X , and the type itself is X . Elimination is non-dependent — the return type X is the same regardless of whether the value is true or false.

$$\text{DBool} := \text{fix}(B \leq \top). \iota(\text{self} \leq B). \lambda(P \leq B \rightarrow \top). \lambda(t \leq P \text{ true}). \lambda(f \leq P \text{ false}). P \text{ self}$$

DBool is the *dependent* counterpart: its motive P is a function from values to types, so the return type $P \text{ self}$ varies with the value being eliminated. This is what ι enables — the self binder in $\iota(\text{self} \leq B)$ refers to the actual value inhabiting the type, and [S-Iota-Intro] substitutes it into $P \text{ self}$ during type checking.

Crucially, Bool and DBool *share the same constructors* {true, false}. The dependent elimination machinery lives entirely in the type (DBool’s ι and motive), not in the constructors. At runtime, a DBool value is identical to a Bool value — the difference is purely compile-time. This pattern recurs later with Fin reusing Nat’s constructors (Section 5.4).

5.2 Pairs

$$\begin{aligned}
\text{Pair } A \ B &:= \lambda(X \leq \top). \lambda(k \leq A \rightarrow B \rightarrow X). X \\
\text{pair } A \ B \ a \ b &:= \lambda(X \leq \top). \lambda(k \leq A \rightarrow B \rightarrow X). k \ a \ b
\end{aligned}$$

Church-encoded binary products. Pair projections fst/snd erase the unused component’s type to $\top$, so call sites only supply the type of the component being projected.

5.3 Natural numbers

$$\begin{aligned}
\text{zero} &:= \lambda(P \leq \top). \lambda(z \leq \top). \lambda(s \leq \top). z \\
\text{succ } n &:= \lambda(n \leq \top). \lambda(P \leq \top). \lambda(z \leq \top). \lambda(s \leq n \rightarrow \top). s \ n \\
\text{Nat} &:= \text{fix}(N \leq \top). \iota(\text{self} \leq N). \\
&\quad \lambda(P \leq N \rightarrow \top). \lambda(z \leq P \ \text{zero}). \\
&\quad \lambda(s \leq \lambda(n \leq N). P \ (\text{succ } n)). P \ \text{self}
\end{aligned}$$

Nat uses both key features of Och. The outer `fix` ties the recursive knot (Nat refers to itself in the motive domain). The inner `ι` binds `self` to the value being typed, enabling *dependent* elimination: the return type `P self` varies with the value.

The constructors are plain lambdas with $\top$ domains — they carry no reference to Nat, so `zero` $\sqsubseteq$ Nat and `succ n` $\sqsubseteq$ Nat hold by the coinductive seen-set discipline (productive unfold of `fix/ι`, then all contravariant domain checks discharge via [S-Top]).

Addition is defined with an explicit `fix` since the Scott-style eliminator provides no induction hypothesis:

$$\begin{aligned}
\text{add} &:= \text{fix } \text{add} \leq \text{Nat} \rightarrow \text{Nat} \rightarrow \text{Nat}. \\
&\quad \lambda(n \leq \text{Nat}). \lambda(m \leq \text{Nat}). \\
&\quad n \ (\lambda(_ \leq \text{Nat}). \text{Nat}) \ m \ (\lambda(\text{pred} \leq \text{Nat}). \text{succ } (\text{add } \text{pred } m))
\end{aligned}$$

5.4 Finite sets

$$\begin{aligned}
\text{Fin} &:= \text{fix}(F \leq \text{Nat} \rightarrow \top). \lambda(n \leq \text{Nat}). \\
&\quad n \ (\lambda(_ \leq \text{Nat}). \top) \ \perp \\
&\quad (\lambda(\text{pred} \leq \text{Nat}). \iota(\text{self} \leq \text{Nat}). \\
&\quad \quad \lambda(P \leq \text{Nat} \rightarrow \top). \lambda(\text{fz} \leq P \ \text{zero}). \\
&\quad \quad \lambda(\text{fs} \leq \lambda(q \leq F \ \text{pred}). P \ (\text{succ } q)). P \ \text{self})
\end{aligned}$$

Fin `n` is the type of naturals strictly less than `n`. The zero-length case is $\perp$ (primitive bottom), so `Fin zero` is uninhabited. The successor case is an ι -type whose `self` is a Nat — this means every `Fin` value is automatically a `Nat` value (`Fin n` $\sqsubseteq$ Nat). **(I consider this a majorly cool result)**

Natural number literals inhabit `Fin` by subsumption: `zero` $\sqsubseteq$ `Fin (succ n)` holds for any `n`, and `n` $\sqsubseteq$ `Fin n` is correctly rejected (the diagonal). No separate FZ/FS constructors are needed.

5.5 Length-indexed arrays

$$\begin{aligned}
\text{Array} &:= \text{fix}(\text{Arr} \leq \text{Nat} \rightarrow \top \rightarrow \top). \lambda(n \leq \text{Nat}). \lambda(T \leq \top). \\
&\quad n \ (\lambda(_ \leq \text{Nat}). \top) \ \text{Unit} \ (\lambda(\text{pred} \leq \text{Nat}). \text{Pair } T \ (\text{Arr } \text{pred } T))
\end{aligned}$$

Array `n T` computes by eliminating the length index: `Array zero T` $\equiv$ `Unit` and `Array (succ k) T` $\equiv$ `Pair T (Array k T)`. Arrays are built with `unit/pair` directly.

Vectors package a length with an array of that length, using a dependent pair (`Sigma`):

$$\text{Vec } T := \text{Sigma } \text{Nat} \ (\lambda(n \leq \text{Nat}). \text{Array } n \ T)$$

5.6 appendVec: catching the wrong addition

The north-star example. `appendVec` unpacks two vectors, concatenates their arrays with `appendArrays`, and repacks the result with the summed length:

$$\begin{aligned}
\text{appendVec} &:= \lambda(T \leq \top). \lambda(v_1 \leq \text{Vec } T). \lambda(v_2 \leq \text{Vec } T). \\
&\quad v_1 \ (\text{Vec } T) \ (\lambda(n_1 \leq \text{Nat}). \lambda(\text{arr}_1 \leq \text{Array } n_1 \ T). \\
&\quad \quad v_2 \ (\text{Vec } T) \ (\lambda(n_2 \leq \text{Nat}). \lambda(\text{arr}_2 \leq \text{Array } n_2 \ T). \\
&\quad \quad \text{mkVec } T \ (\text{add } n_1 \ n_2) \ (\text{appendArrays } T \ n_1 \ n_2 \ \text{arr}_1 \ \text{arr}_2)))
\end{aligned}$$

This type-checks: the length add $n_1\ n_2$ matches the length of the concatenated array.

Now consider a version with a deliberate bug — add $n_1\ n_1$ instead of add $n_1\ n_2$:

$$\text{appendVec}_{\text{wrong}} := \dots \text{mkVec } T \text{ (add } n_1\ n_1) \text{ (appendArrays } T\ n_1\ n_2\ \text{arr}_1\ \text{arr}_2) \dots$$

The type checker *rejects* this: $\text{arr}_2 \sqsubseteq \text{Array } n_1\ T$ fails because arr_2 has type $\text{Array } n_2\ T$ and $n_2 \neq n_1$. The system catches a bug as subtle as adding the wrong two numbers.

5.7 indexArr: safe array indexing

`indexArr` looks up the i -th element of an $\text{Array } n\ T$, where i has type $\text{Fin } n$:

$$\begin{aligned} \text{indexArr} := & \text{fix } (\text{self} \leq (\lambda(T \leq \top). \lambda(n \leq \text{Nat}). \text{Array } n\ T \rightarrow \text{Fin } n \rightarrow T)). \\ & \lambda(T \leq \top). \lambda(n \leq \text{Nat}). n \ (\lambda(m \leq \text{Nat}). \text{Array } m\ T \rightarrow \text{Fin } m \rightarrow T) \\ & (\lambda(\text{arr} \leq \text{Array } \text{zero } T). \lambda(i \leq \text{Fin } \text{zero}). i) \\ & (\lambda(p \leq \text{Nat}). \lambda(\text{arr} \leq \text{Array } (\text{succ } p)\ T). \lambda(i \leq \text{Fin } (\text{succ } p)). \dots) \end{aligned}$$

In the zero-length branch, i has type $\text{Fin } \text{zero} = \perp$, so the branch body is i itself — *ex falso* via [S-BotL]. In the successor branch, the array is destructured as a pair and i eliminates to either the head or a recursive call on the tail.

The payoff is at call sites:

$$\begin{aligned} \text{indexArr Nat three arr zero} & \quad \checkmark \quad (\text{zero} \sqsubseteq \text{Fin three}) \\ \text{indexArr Nat three arr two} & \quad \checkmark \quad (\text{two} \sqsubseteq \text{Fin three}) \\ \text{indexArr Nat three arr three} & \quad \times \quad (\text{three} \not\sqsubseteq \text{Fin three}) \end{aligned}$$

Out-of-bounds access is a *compile-time* error: the index literal fails to subtype $\text{Fin } n$, and no runtime check is needed.

6 Metatheory

Soundness connects the concrete semantics (Section 3.1), subtyping (Section 4.4), and well-formedness (Section 4.2).

Evaluation equivalence. If e evaluates to e' , both directions of subtyping hold.

$$e \Downarrow e' \text{ implies } \emptyset; \emptyset \vdash e' \sqsubseteq e \text{ and } \emptyset; \emptyset \vdash e \sqsubseteq e'$$

Progress. A well-formed closed term doesn't get stuck during evaluation. The evaluator may return a value or run out of fuel but never reaches an error state.

End-to-end. Composing the above: if e is well-formed and subtypes τ , and e evaluates to e' , the result subtypes τ .

$$\emptyset \vdash e \text{ wf and } \emptyset; \emptyset \vdash e \sqsubseteq \tau \text{ and } e \Downarrow e' \text{ implies } \emptyset; \emptyset \vdash e' \sqsubseteq \tau$$

6.1 What soundness promises to the programmer

Two separable runtime properties a type system can promise:

1. **Termination.** Running a well-typed program eventually produces a value. A totality / normalization claim.
2. **No runtime type errors.** Conditional on the program *not* diverging, the value it produces is well-formed and the program never reaches a stuck state.

Och **does not aim to prove (1)**. Consistency and normalization are explicitly deferred; the calculus admits non-terminating terms by design (type-in-type; general recursion via `fix`).

Och **does aim to prove (2)**, and this is the real runtime guarantee of the type system. The language expects the following discipline: **type-check first; only evaluate if it succeeded**.

7 Lean Formalisation

The calculus described above has been formalised in Lean 4, which is used to maintain a test suite of example programs and allow AI agents to iterate on the metatheory. The source is available at github.com/charlielidbury/ochre.

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